House Price Analysis and Feature Exploration

EDA Assignment 2

Ames Housing · California Housing · House Sales in King County

Tools: Python, pandas, matplotlib, seaborn, scikit-learn

1. Objective and summary

This study explores what drives residential house prices in three well-known datasets, cleans each one, visualises the main relationships and then fits predictive models to measure how well – and through which features – prices can be explained.

- **Ames (Iowa):** price is driven by build quality and size. Overall quality has r = 0.81 and the engineered *total_sf* feature has the highest correlation of all. Regularised linear regression is best (test $R^2 = 0.96$).
- **King County (Washington):** location dominates. Latitude is the top feature, then living area and grade. Waterfront homes sell for about 3.1× the median of non-waterfront homes. Gradient Boosting ($R^2 = 0.91$) far outperforms linear models ($R^2 \approx 0.70$).
- **California:** latitude, longitude and median income lead. Non-linear models ($R^2 = 0.85$) clearly beat linear ones ($R^2 = 0.57$). Values are capped at $500k, which limits accuracy at the top end.
- **Common lessons:** price is right-skewed, so log-transforming it helps; engineered features (total area, age, ratios) are among the most useful predictors; geography needs a non-linear model.

2. Data

Dataset	Rows	Columns	Period / source	Target
Ames Housing	2,930	82	Ames Assessor, 2006–2010	SalePrice
King County	21,613	21	May 2014 – May 2015	price
California Housing	20,640	10	1990 US census block groups	median_house_value

3. Data cleaning

Each dataset was inspected for missing values, duplicates, impossible values and outliers. Decisions were made from the meaning of each field rather than by blanket deletion.

Dataset	Rows	Cols	Main actions
Ames	2,930 → 2,925	82 → 92	'NA' in Alley, Pool QC, Fence, Fireplace, basement and garage fields means *feature absent* → 'None'. Numeric gaps → 0. Lot Frontage → neighbourhood median (+flag). Garage Yr Blt fixed. 5 homes >4,000 sq ft removed. Added house_age, total_sf, total_bath, has_* flags, log price.
King County	21,613 → 21,603	21 → 30	Date parsed; zipcode as text; one 33-bedroom typo fixed to 3; 10 rows with 0 bathrooms removed. Added house_age, renovated, price_per_sqft, repeat_sale, log price.
California	20,640 → 20,561	10 → 18	207 missing total_bedrooms imputed from median bedroom/room ratio (+flag); cap flags for price, age, income; 79 implausible blocks removed; added per-household ratios.

4. Exploratory analysis

4.1 Ames Housing

Sale price is right-skewed (skew 1.59) and nearly symmetric after a log transform (–0.04). Quality and size are the strongest drivers, and neighbourhood creates large price tiers (StoneBr, NridgHt and NoRidge are the most expensive, with medians around $300k–$320k). Newer homes sell for more, shown by the negative correlation with house age.

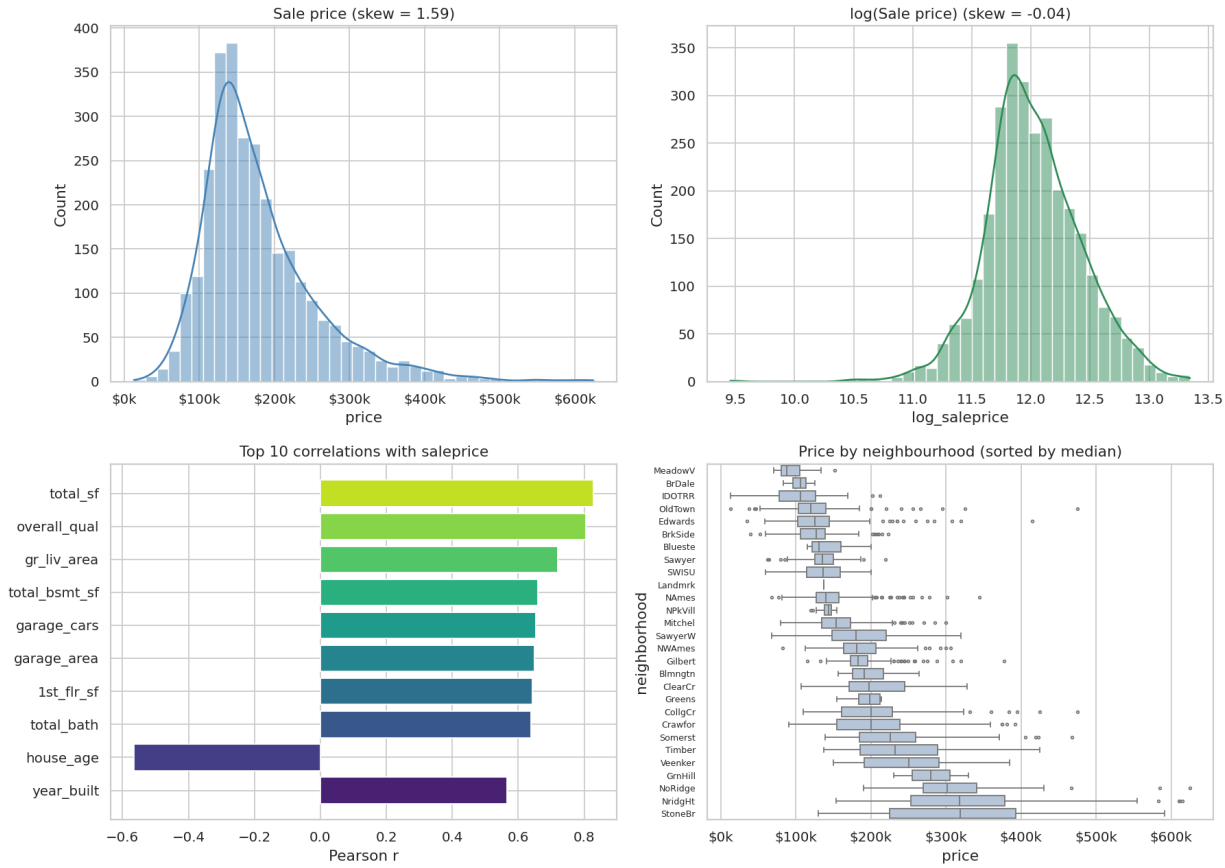


Figure 1 – Ames: price distribution, top correlations and price by neighbourhood

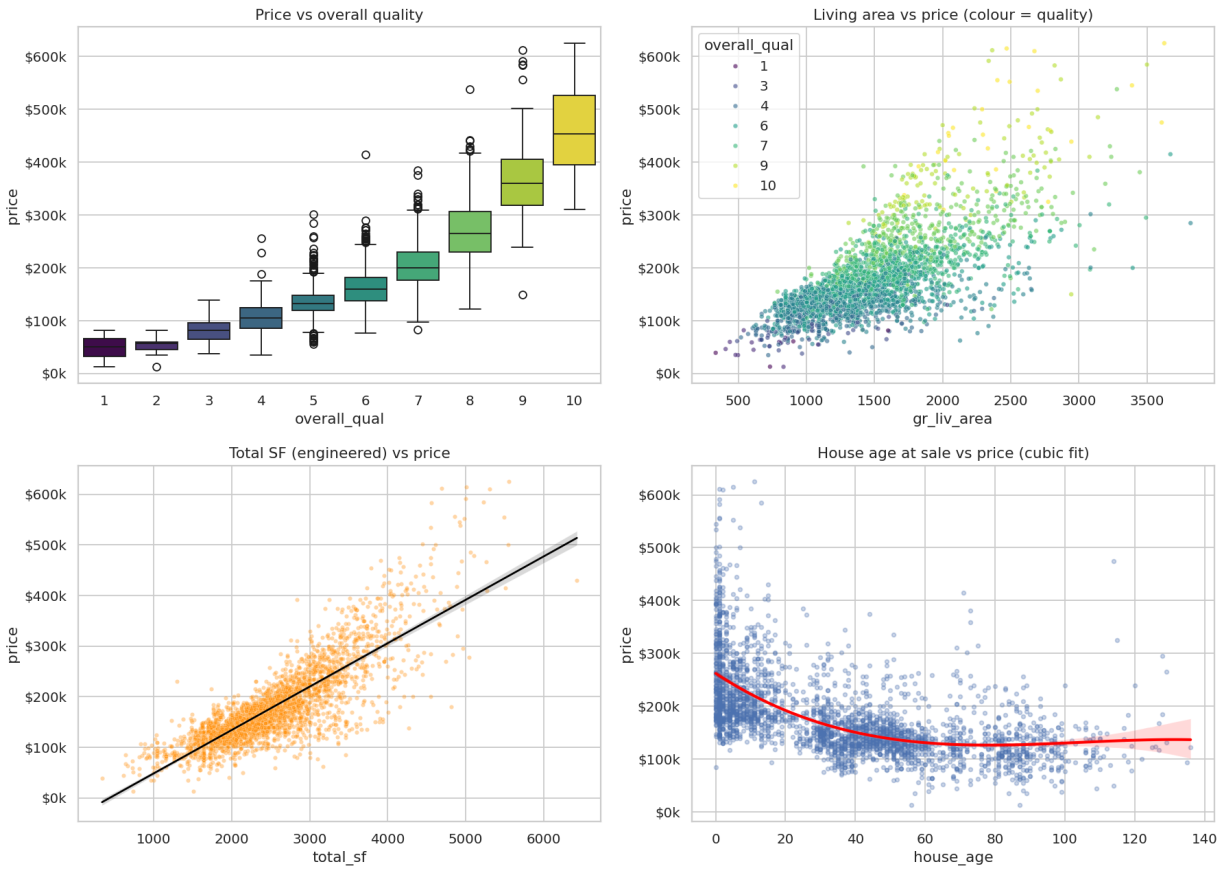


Figure 2 – Ames: quality, living area, total area and age versus price

4.2 King County

Living area (r = 0.70) and grade (r = 0.67) have the strongest linear relationship with price, but the map shows how strongly location matters: the Seattle–Bellevue corridor and the lakefront are the most expensive, and zipcode 98039 has a median above $1.8M. Waterfront properties have a median price about 3.1× that of other homes. Monthly median price peaks in spring 2015 (the final month is a partial month).

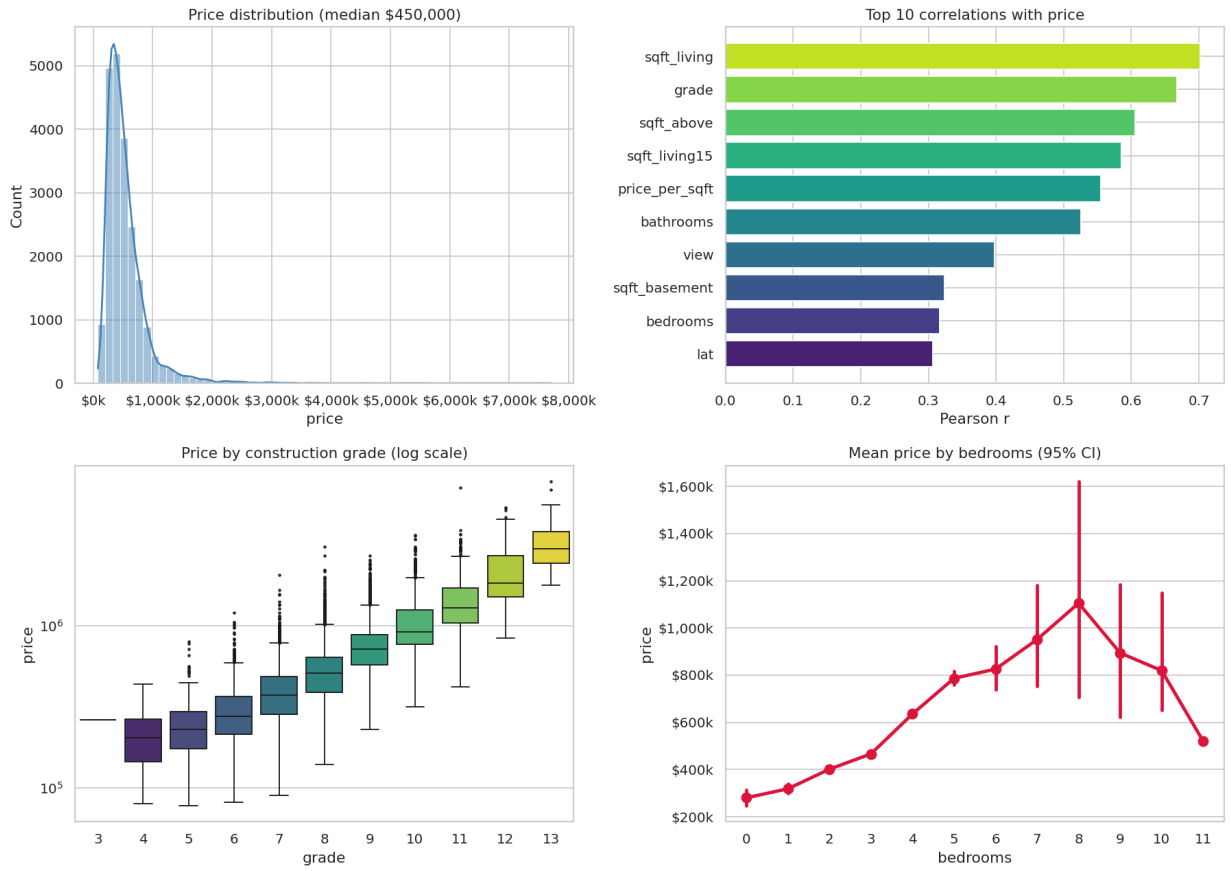


Figure 3 – King County: price distribution, correlations, grade and bedrooms

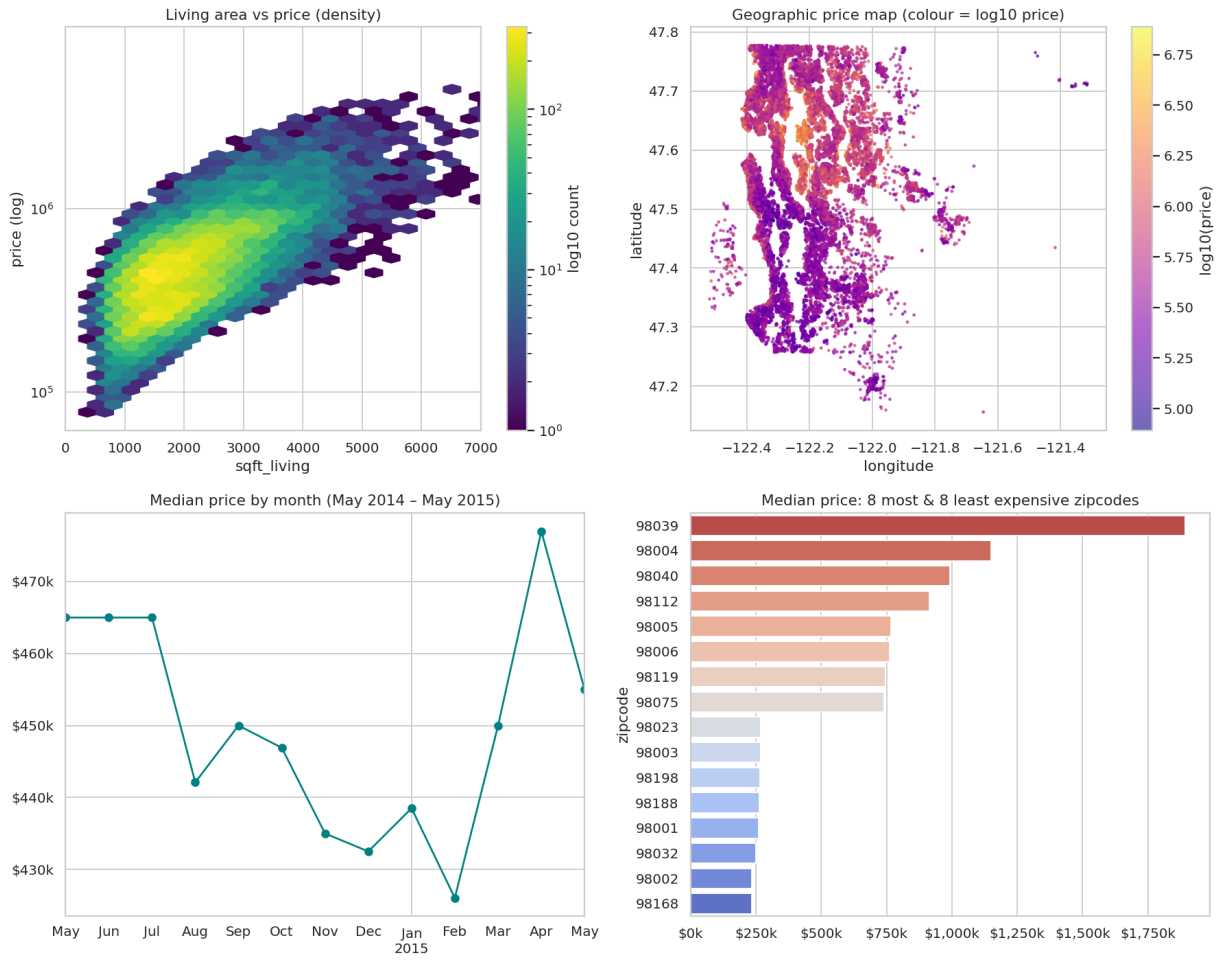


Figure 4 – King County: size, location, time trend and zipcode ranking

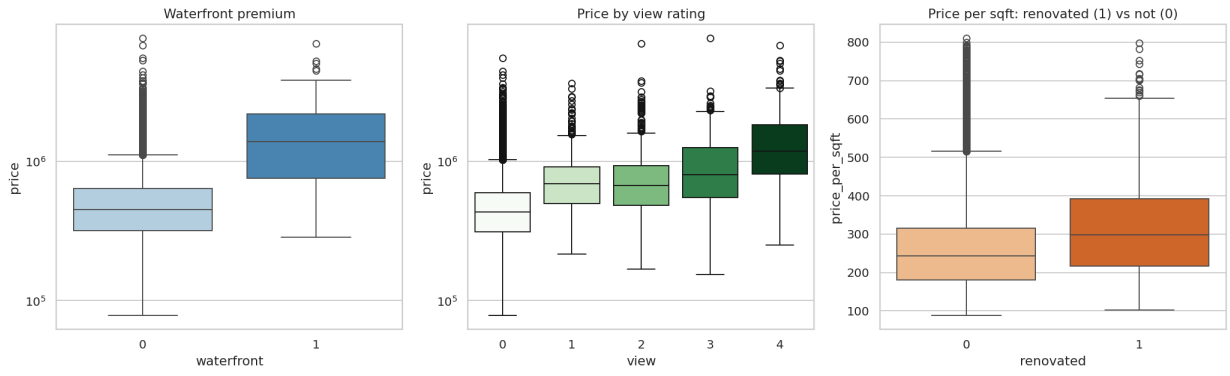


Figure 5 – King County: waterfront, view and renovation premiums

4.3 California Housing

Median income is the strongest linear predictor (r = 0.69). The map shows an expensive coastal belt around the Bay Area and Los Angeles; inland blocks have a median of about $108k versus $215k–$234k near the coast or bay. The pile-up at $500k is a data cap, so the cap was flagged rather than removed.

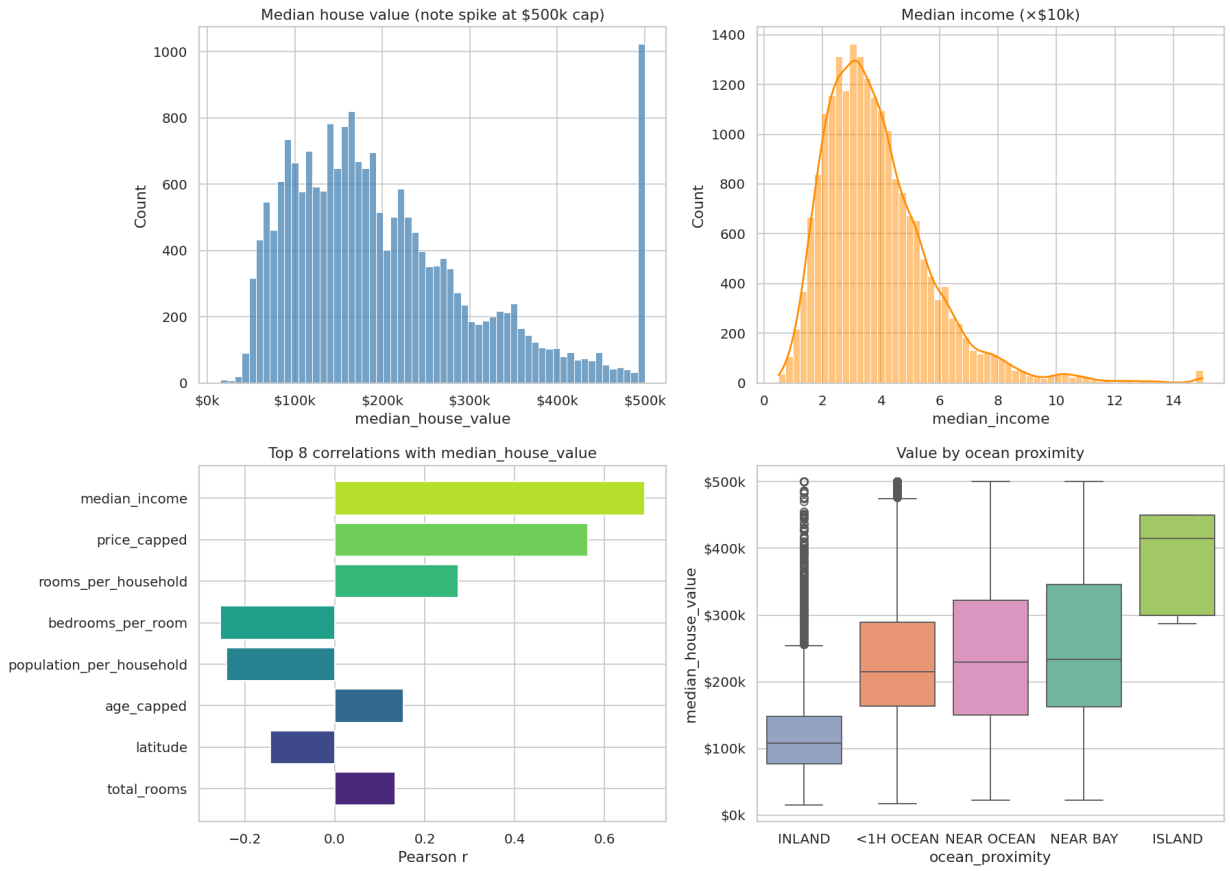


Figure 6 – California: value distribution, income, correlations and ocean proximity

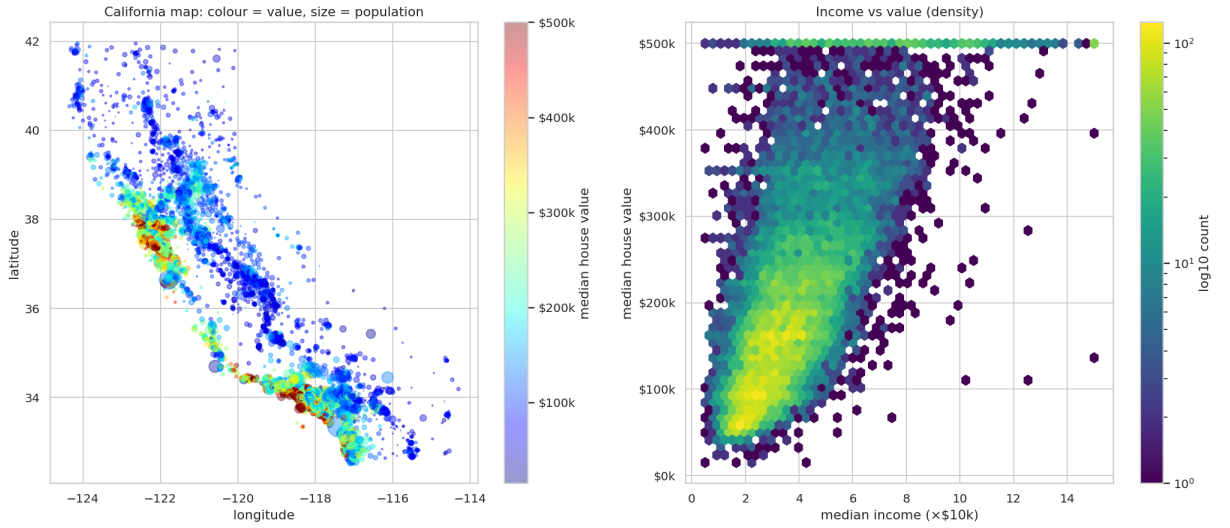


Figure 7 – California: geographic map and income-versus-value density

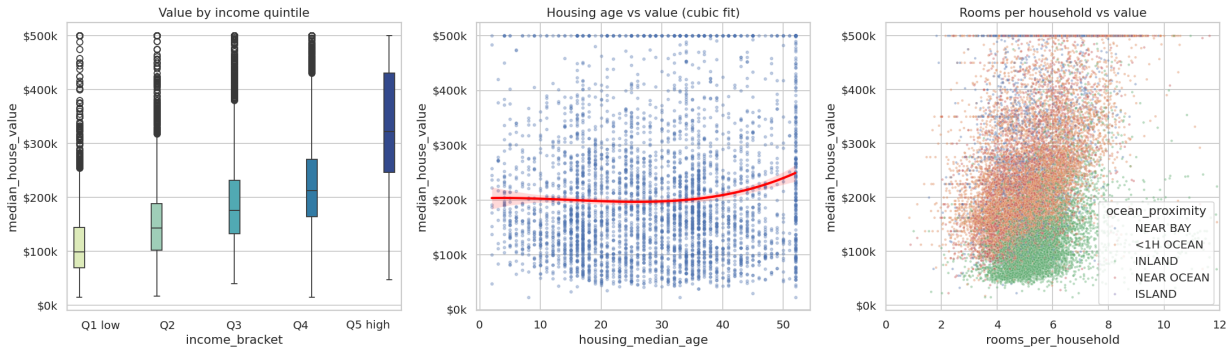


Figure 8 – California: income quintiles, housing age and rooms per household

4.4 Correlation structure

Heatmaps show strong multicollinearity between the size variables (living area, above-ground area, total area, garage and basement measures). This is why regularised regression (Ridge) works well on Ames, and why tree models are more robust for King County and California.

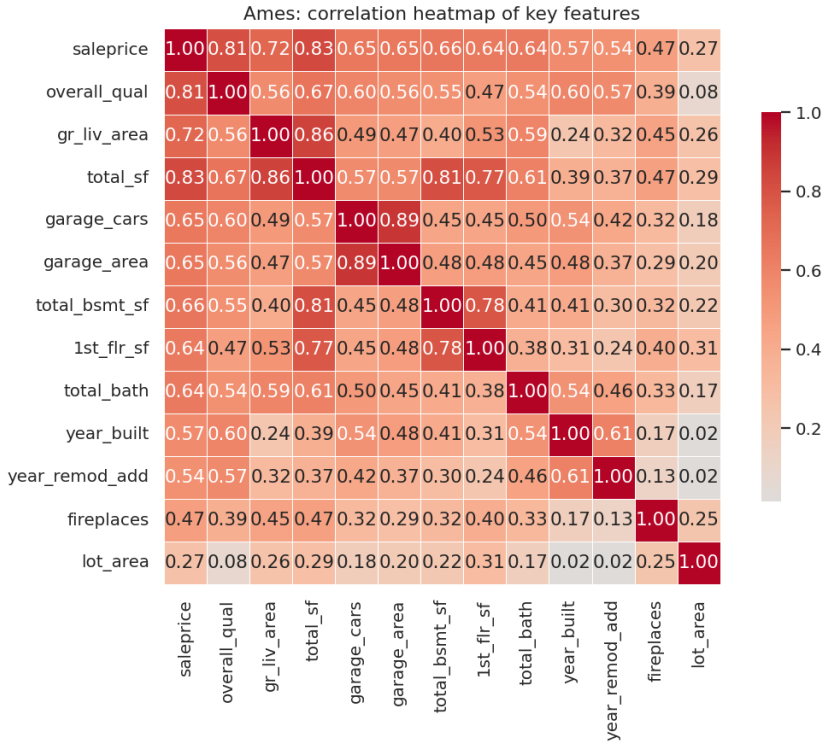


Figure 9 – Ames correlation heatmap

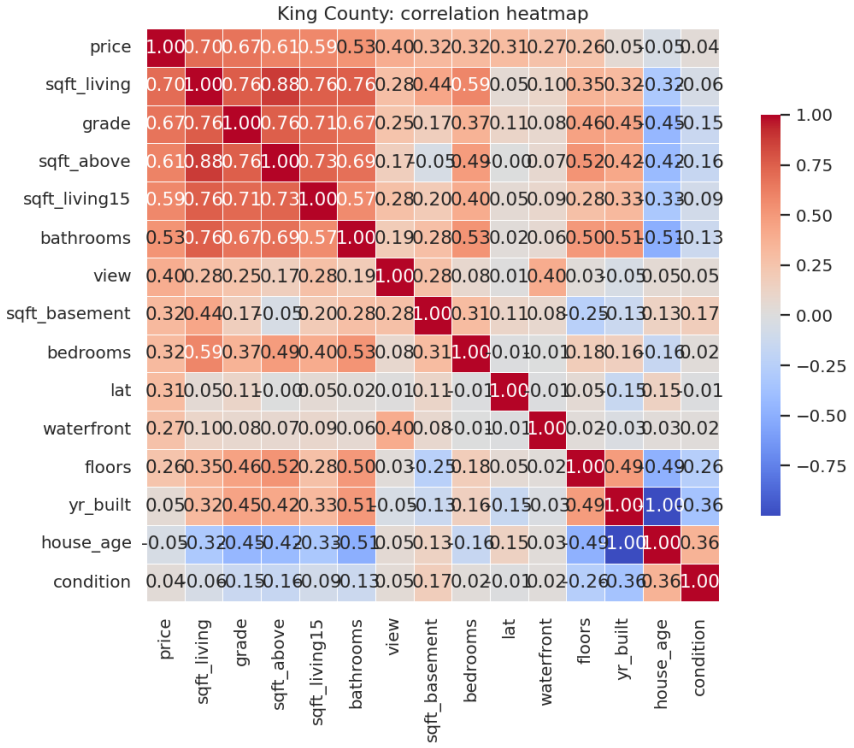


Figure 10 – King County correlation heatmap

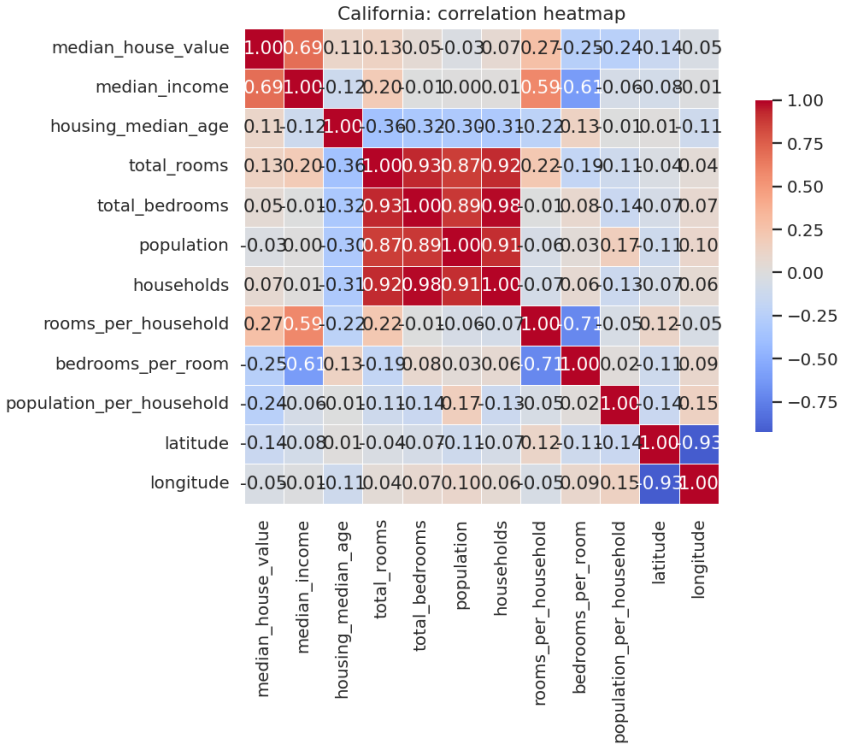


Figure 11 – California correlation heatmap

5. Modelling

Four models were compared on each dataset: Linear Regression, Ridge, Random Forest and Gradient Boosting. The target is log(price); data were split 80/20 (random_state 42) and 3-fold cross-validation was run on the training portion. Errors are reported in dollars. Price-derived columns (price_per_sqft, the cap flag and the log target) were excluded from the predictors to prevent leakage. The best model per dataset is highlighted.

Dataset	Model	CV R^2 (log)	Test R^2 (log)	Test R^2 ($)	RMSE ($)	MAE ($)
Ames	Linear Regression	0.892	0.953	0.958	17,326	12,038
Ames	**Ridge (alpha=10)**	**0.903**	**0.956**	**0.959**	**17,182**	**11,748**
Ames	Random Forest	0.883	0.933	0.932	21,906	13,726
Ames	Gradient Boosting	0.891	0.941	0.945	19,838	12,646
King County	Linear Regression	0.880	0.880	0.691	207,264	78,343
King County	Ridge (alpha=10)	0.878	0.878	0.696	205,558	79,320
King County	Random Forest	0.882	0.890	0.858	140,694	71,752
King County	**Gradient Boosting**	**0.906**	**0.911**	**0.906**	**114,387**	**62,832**
California	Linear Regression	0.693	0.691	0.574	76,205	48,490
California	Ridge (alpha=10)	0.693	0.691	0.574	76,206	48,496
California	Random Forest	0.826	0.844	0.815	50,225	31,993
California	**Gradient Boosting**	**0.849**	**0.867**	**0.845**	**45,959**	**28,789**

- **Ames:** Ridge is best (R^2 = 0.959 in dollars, RMSE ≈ $17k). With strong engineered features and only about 2,900 rows, a regularised linear model generalises better than the trees.
- **King County:** Gradient Boosting reaches R^2 = 0.906 and RMSE ≈ $114k, against 0.69–0.70 for the linear models. The gap shows how non-linear the location effect is.
- **California:** Gradient Boosting reaches R^2 = 0.845 and RMSE ≈ $46k, against 0.57 for linear models. Latitude and longitude need interactions that trees capture naturally.

6. Feature importance

Permutation importance (the drop in test R^2 when one feature is shuffled) was computed for the best model in each dataset. The full list is in feature_importance.csv.

Ames (Ridge (alpha=10))

Rank	Feature	Import.
1	overall_qual	0.077
2	gr_liv_area	0.038
3	total_sf	0.028
4	1st_flr_sf	0.019
5	overall_cond	0.016
6	neighborhood	0.015
7	2nd_flr_sf	0.013
8	house_age	0.007

King County (Gradient Boosting)

Rank	Feature	Import.
1	lat	0.615
2	sqft_living	0.179
3	grade	0.129
4	long	0.063
5	sqft_lot	0.033
6	zipcode	0.015
7	sqft_living15	0.015
8	view	0.014

California (Gradient Boosting)

Rank	Feature	Import.
1	latitude	0.484
2	longitude	0.396
3	median_income	0.311
4	ocean_proximity	0.178
5	population_per_household	0.101
6	bedrooms_per_room	0.027
7	housing_median_age	0.026
8	rooms_per_household	0.024

- Ames: **overall_qual** is the single most important feature, followed by living area and total area.
- King County: **lat** alone accounts for far more predictive power than any other feature, then sqft_living, grade and long. Price is about where the house is.
- California: **latitude, longitude, median_income** and ocean_proximity dominate; the engineered population_per_household is also in the top five.

7. Conclusions and limitations

- Quality and size matter everywhere, but location is the deciding factor in the two metropolitan datasets.
- Cleaning choices matter: treating ‘NA’ as ‘feature absent’ in Ames, flagging rather than dropping capped values in California, and fixing the 33-bedroom typo in King County.
- Log-transforming the target and engineering simple features (total area, age, ratios) improved every dataset.
- Limitations: the datasets cover different places and years, so prices are not directly comparable; California is block-group-level data with a $500k cap; King County has no data past May 2015; permutation importance is shared among correlated features, so individual values should be read as indicative.
- Possible next steps: hyper-parameter tuning, spatial features (distance to downtown or coast), and a model that handles capped targets.